

MAE 3128

Biomechanics-I



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Topics covered today:

1. Deformations due to Torsion
 1. Anatomy of the long bone
 2. Biological motivations and applications
2. Stress due to torsion
3. In-class problems



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Summary of what's been covered in this course so far

- **Lectures 1, 2:** Introduced **Biomechanics** and reviewed **Statics** under rigid body assumptions
- **Lectures 3, 4:** Introduced **general concepts of stress and strain**, and **constitutive relationship using Hooke's law** for small-strain, solid-like materials.
- **Lectures 4, 5:** Shift focus to **canonical problems** in introductory biosolid mechanics
 - **For each problem,**
 - Choose coordinate systems
 - Calculate **stress components** by applying **equilibrium equations** (force, linear and angular momentum balances without inertia).
 - **Stress** is expressed in terms of **applied load and geometry**
 - **Material properties** appearing explicitly only in a few cases.
 - Calculate **strain or deformation**, depending on **load, geometry, and material properties**.

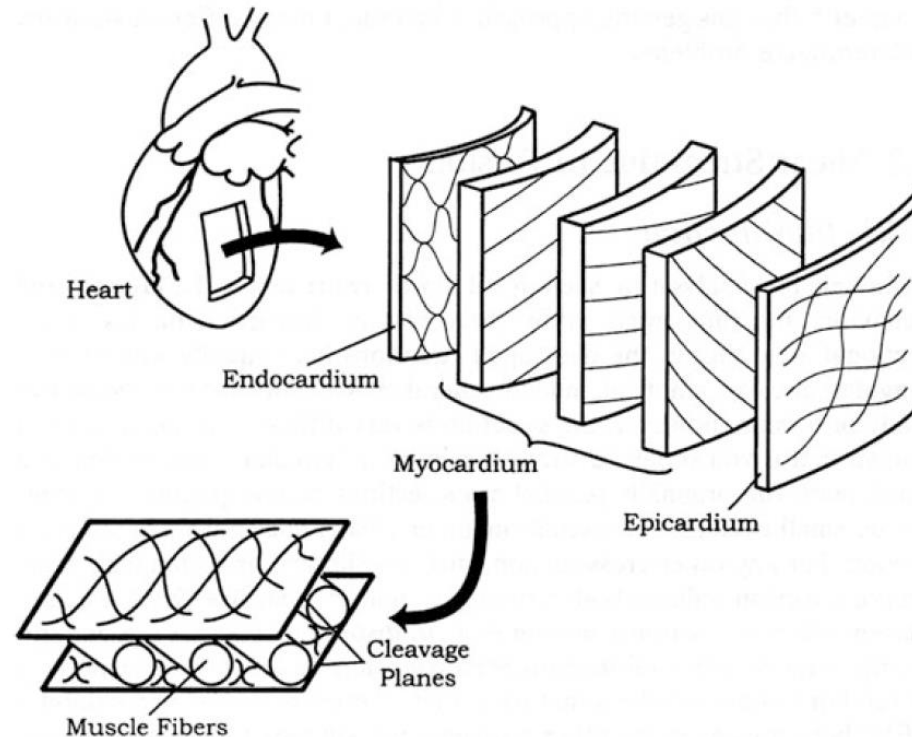
After the midterm exam the following **canonical problems** will be addressed:

- **Extension of rods and torsion of cylindrical structures**
- **Bending of beams**
- **Buckling of columns**



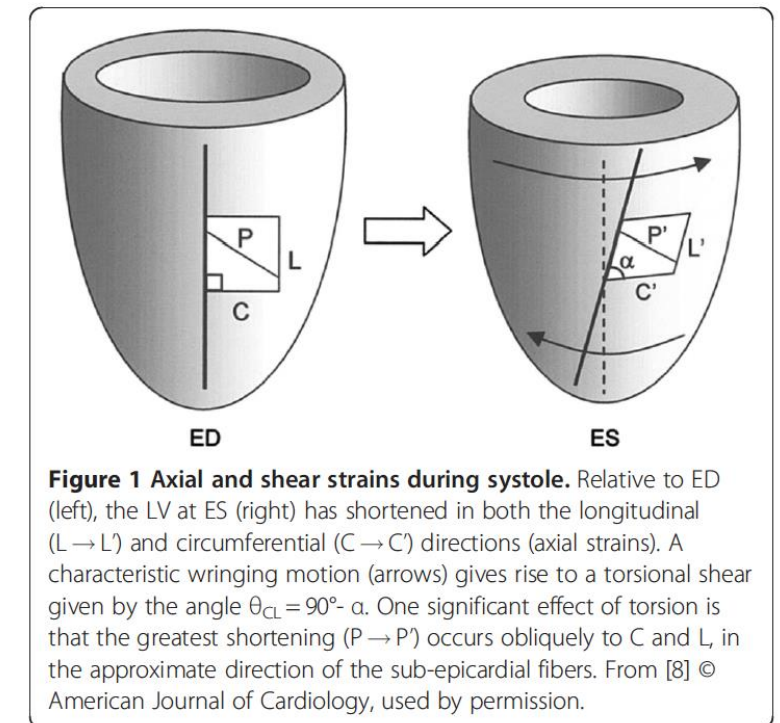
Biological Motivation: Why Torsion Matters in the Heart?

- Many tissues and implants experience twisting (**torsion**) under load.
- The heart's twist is essential for efficient blood ejection each beat.
- Contraction combines shortening, constriction, twisting, and shearing of cardiac muscle.



References:

1. Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)
2. Eric J. Stöhr, Rob E. Shave, Aaron L. Baggish, and Rory B. Weiner, Left ventricular twist mechanics in the context of normal physiology and cardiovascular disease: a review of studies using speckle tracking echocardiography, *American Journal of Physiology-Heart and Circulatory Physiology* 2016 311:3, H633-H644
3. Young AA, Cowan BR. Evaluation of left ventricular torsion by cardiovascular magnetic resonance. *J Cardiovasc Magn Reson.* 2012 Jul 24;14(1):49. doi: 10.1186/1532-429X-14-49. PMID: 22827856; PMCID: PMC3461493.



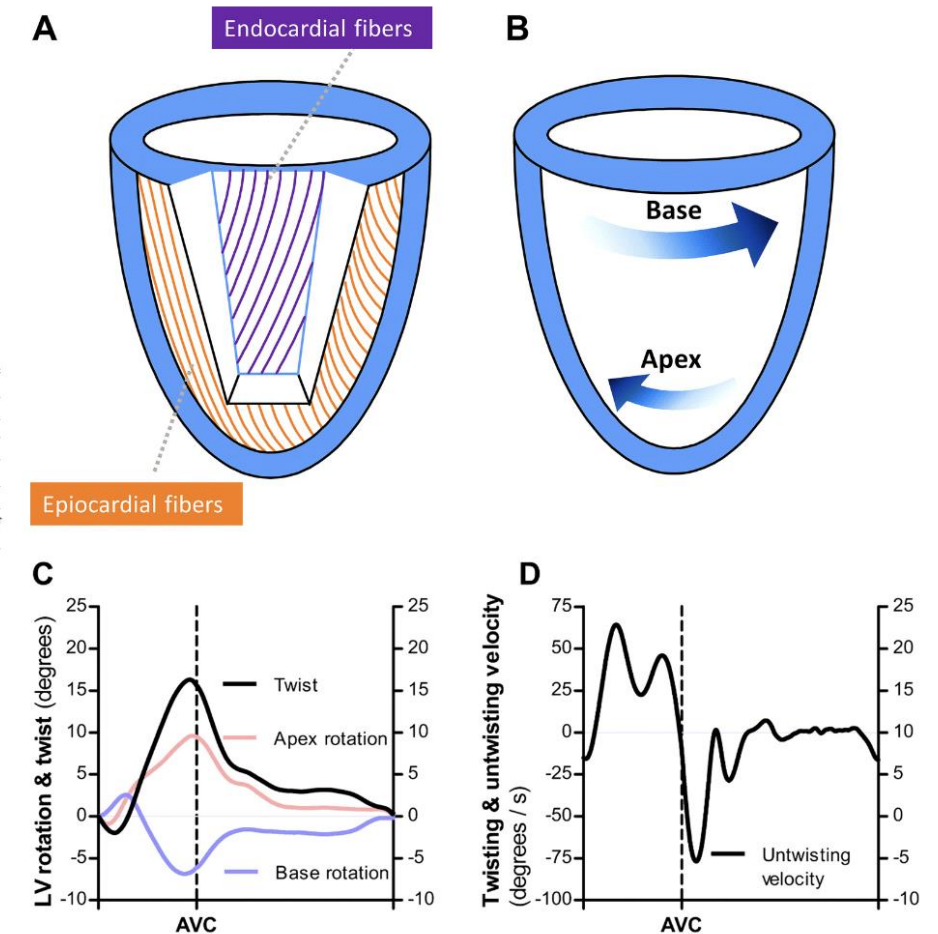
Biological Motivation: Why Torsion Matters in the Heart?

- A unique helical fiber architecture drives this twisting motion.
- Biomechanical modeling in the 1970s–80s revealed how fiber orientation enables twist.
- Twisting not only pumps blood but also evens out stress across the ventricular wall.
- More uniform stress means cardiac cells see similar, potentially optimal mechanical cues.

References:

1. Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)
2. Eric J. Stöhr, Rob E. Shave, Aaron L. Baggish, and Rory B. Weiner, Left ventricular twist mechanics in the context of normal physiology and cardiovascular disease: a review of studies using speckle tracking echocardiography, *American Journal of Physiology-Heart and Circulatory Physiology* 2016 311:3, H633-H644

Fig. 1. Helices and twist in the left ventricle (LV). A: LV endocardial and epicardial helix configuration in end diastole. B: rotational directions of the LV base and apex. C: typical LV rotation and twist in a healthy individual during resting conditions. D: typical untwisting velocity in a healthy individual during resting conditions. For the purpose of clarity the underlying base and apex velocities have been omitted.



Biological Motivation: Twisting moments in Femoral implants

Reference: Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)

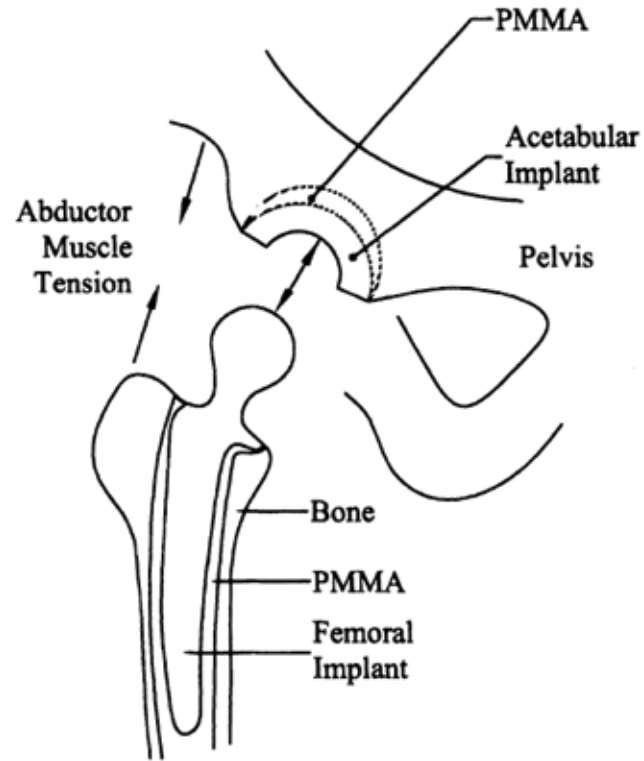


FIGURE 4.10 Schema of the femur and acetabulum, a cup-shaped cavity in which the head of the femur articulates. Because the line of action of the loads applied on the femoral head do not coincide with the long axis of the mid-shaft of the femur, these forces can cause both bending and twisting moments in addition to axial compression.

Loads acting on the hip

Complex (compressive, bending, and twisting) loads occur naturally during daily activities.



Review of some basics

Torque: Rotational Force

Definition

Torque (T) is the rotational equivalent of linear force; the tendency of a force to rotate an object about an axis.

$$T = rF\sin(\theta)$$

- T = Torque (N·m)
- r = Lever arm distance (m)
- F = Applied force (N)
- θ = Angle between r and F

Hookean LEHI model

$$\begin{aligned}\epsilon_{rr} &= \frac{1}{E}[\sigma_{rr} - \nu(\sigma_{\theta\theta} + \sigma_{zz})] + \beta\Delta T, & \epsilon_{r\theta} &= \frac{1}{2G}\sigma_{r\theta}, \\ \epsilon_{\theta\theta} &= \frac{1}{E}[\sigma_{\theta\theta} - \nu(\sigma_{rr} + \sigma_{zz})] + \beta\Delta T, & \epsilon_{rz} &= \frac{1}{2G}\sigma_{rz}, \\ \epsilon_{zz} &= \frac{1}{E}[\sigma_{zz} - \nu(\sigma_{rr} + \sigma_{\theta\theta})] + \beta\Delta T, & \epsilon_{\theta z} &= \frac{1}{2G}\sigma_{\theta z}.\end{aligned}$$

